Lesson 8 - Trees

8.1 Properties of trees

Question 111

Level of difficulty: E

How many edges does a tree with n nodes have? Explain.

Question 112

Level of difficulty: E

A tree with n nodes has exactly n–1 edges. What can be derived from this?

Solution question 111

Solution hint: Successive picking of leaves

Short solution: n–1 edges, successive leaf picking technique.

Detailed solution: Each tree with n nodes has exactly n–1 edges. This can be traced using the technique of successive leaf picking. Let the number of nodes in the tree under consideration be n. If now the leaf and the corresponding edge are picked off, the result is a tree with one node and one edge less. However, the difference between the number of nodes and the number of edges in the old tree and the corresponding difference in the new tree remain the same. This process of picking off individual leaves is repeated until, after n–1 steps, a tree with one node has been created. This tree has no edge. This means that the difference between the number of nodes and the number of edges is equal to 1. Or, in other words, a tree with n nodes contains exactly n–1 edges.

Solution question 112

Solution hint: Familiarize yourself with the theory about trees.

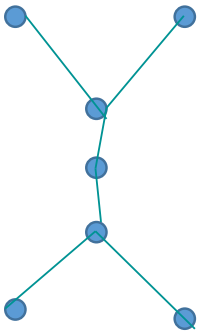
Solution: A graph is a tree if and only if it does not contain a closed walk, but adding exactly one new edge to the existing nodes creates a closed walk.

If a tree has a node of degree k, then it has at least k leaves. A connected graph is a tree exactly if removing any edge makes it no longer connected. A connected graph is a tree exactly if all its edges are bridges.

Question 113

Level of difficulty: M

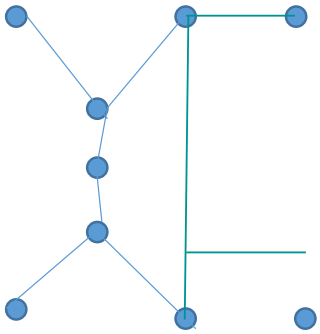
Decide if it is a tree and give reasons for your answer.



Question 114

Level of difficulty: M

Decide if it is a tree and give reasons for your answer.



Solution question 113

Solution hint: A graph with n nodes has n–1 edges.

Short solution: Yes

Detailed solution: The graph contains seven nodes and six edges, so it must be a tree.

Solution question 114

Solution hint: A graph with n nodes has n-1 edges.

Short solution: No

Detailed solution: The graph contains nine nodes and nine edges, so it cannot be a tree. Furthermore, it can be seen that it contains a closed walk.

Question 115

Level of difficulty: D

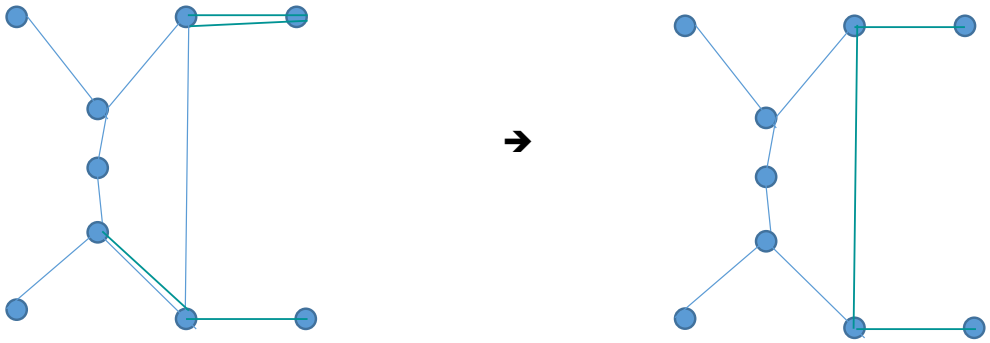
Is the following sentence true? If, from a connected graph with multiple edges, these multiple edges are removed, a tree is formed.

Solution question 115

Solution hint: Familiarize yourself with the requirements for a tree.

Short solution: Not true.

Detailed solution: If, for example, the multiple edges are removed from the graph below, the following graph is obtained:



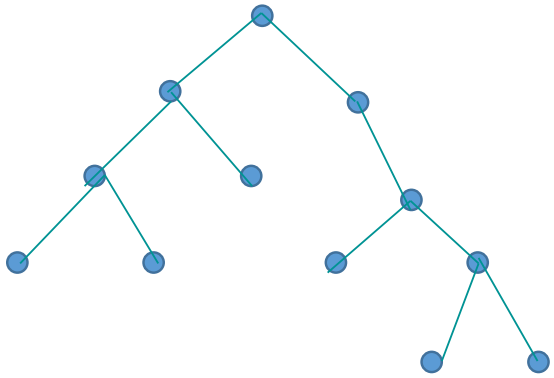
The resulting graph is not a tree.

8.2 Rooted tree

Question 116

Level of difficulty: E

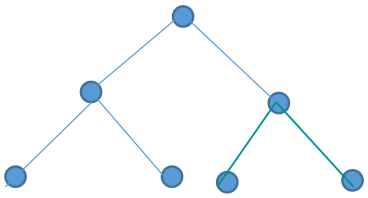
Given is the rooted tree of order 2. Decide whether it is complete and/or filled in. Justify.



Question 117

Level of difficulty: E

Given is the rooted tree of order 2. Decide whether it is complete and/or filled in. Justify.



Solution question 116

Solution hint: For a completed rooted tree, all inner nodes have the highest possible number of children, making the degree either equal to the order of the tree (inner node) or zero (leaf). A complete rooted tree has the highest possible number of nodes at each level if all inner nodes previously had the maximum number of children. Every complete rooted tree is also filled.

Short solution: Not filled in and therefore also not complete

Detailed solution: The given rooted tree is not completed because the node at depth 1 does not have enough children. Thus the rooted tree is also not complete.

Solution question 117

Solution hint: For a completed rooted tree, all inner nodes have the highest possible number of children, making the degree either equal to the order of the tree (inner node) or zero (leaf). A complete root tree has the highest possible number of nodes at each level if all inner nodes previously had the maximum number of children. Every complete rooted tree is also filled.

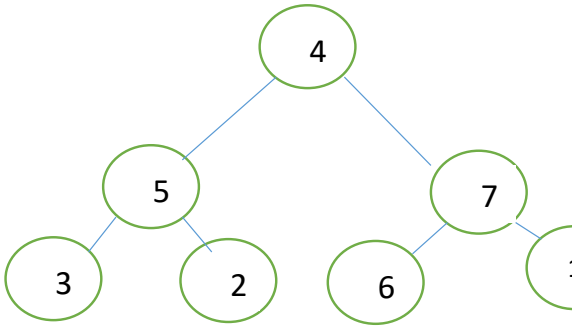
Short solution: Complete and filled in

Detailed solution: It is complete because there are two children at each level and so it is filled in.

Question 118

Level of difficulty: M

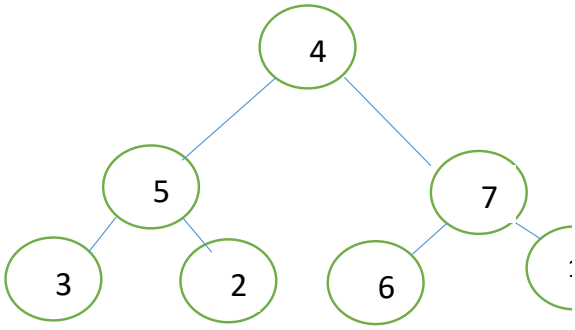
For the following binary search tree, enter the preorder sequence.



Question 119

Level of difficulty: M

For the following binary search tree, enter the inorder sequence.



Solution question 118

Solution hint: Familiarize yourself with the different traversing methods.

Short solution: 4 5 3 2 7 6 1

Detailed solution: First the root, then the left, then the right subtree: 4 5 3 2 7 6 1.

Solution question 119

Solution hint: Familiarize yourself with the different traversing methods.

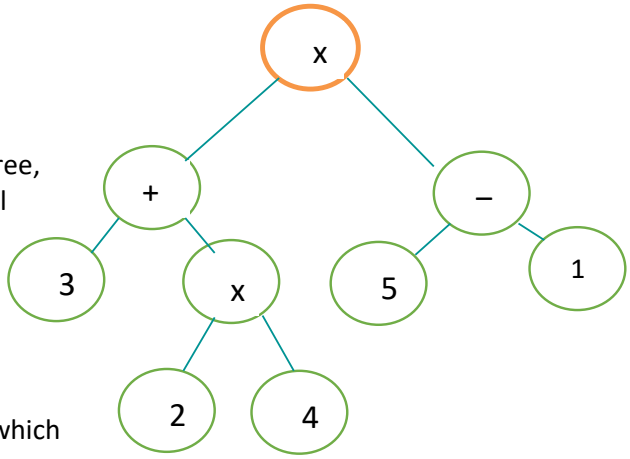
Short solution: 3 5 2 4 6 7 1

Detailed solution: First the left subtree, then the root, then the right subtree: 3 5 2 4 6 7 1.

Question 120

Level of difficulty: D

With a suitable traversing of the syntax tree, it is also possible to use the mathematical expression that is represented in the binary tree. For the given abstract syntax tree:



a) Which flow order is used?

b) What is the mathematical expression which is represented by the syntax tree?

Solution question 120

Solution hint: Familiarize yourself with the traversal.

Short solution: a) Inorder sequence b) 44

Detailed solution: a) The pass order by inorder sequence leads to the mathematical expression, b) $(3 + (2 \cdot 4)) \cdot (5–1) = 44$.

8.3 Spanning tree

Question 121

Level of difficulty: E

What is meant by a tree? Explain how to obtain a tree from any graph that is not one.

Question 122

Level of difficulty: E

How many spanning trees does the complete graph K_5 have?

Solution question 121

Solution hint: Familiarize yourself with the theory of the spanning tree. A tree must not have closed walks.

Short solution: A tree containing all the nodes of a graph; find a closed walk from a graph that is not a tree and eliminate it by removing an edge, and so on.

Detailed solution: A tree containing all nodes of a graph is a spanning tree of the graph. This is created when a closed edge move is picked from any graph that is not a tree and an edge is removed from it, eliminating the closed walk. If there are other closed walks in the graph, another edge is removed from them, further reducing the number of closed walks. In this way, edges are removed until there are no more closed walks and the graph has become a "thinned" tree.

Solution question 122

Solution hint: Cayley's formula

Short solution: 125

Detailed solution: The formula of Cayley (1889) gives an upper limit. It says: A complete graph K_n has n^{n-2} spanning trees. Thus K_5 has 125 spanning trees.

Question 123

Level of difficulty: M

Explain the breadth-first search algorithm for finding spanning trees.

Question 124

Level of difficulty: M

Explain the depth-first search algorithm for finding spanning trees.

Solution question 123

Solution hint: The algorithm traverses all possible walks of length 1 from a node.

Short solution: The algorithm traverses all possible edge paths with length 1 from a node. From the newly reached nodes, it again follows walks of length 1 to unreached nodes. This is continued until there are no more unreached nodes. Since the graph is connected, the algorithm finds a spanning tree after finitely many steps.

Detailed solution:

- Step 1: Set E_0 as the empty set and define two lists L_1 and L_2 also as empty lists. An arbitrary starting node v_1 is chosen and written to lists L_1 and L_2.
- Step 2: If L_1 is empty, then stop and $T = (V, E_0)$ is a spanning tree. Otherwise, take the first node v from L_1.
- Step 3: If all adjacent nodes of v are in L_2, then v is deleted from L_1. Otherwise, choose the "smallest" adjacent node w that is not in L_2 and write it to the end of the lists L_1 and L_2. The edge $\{v, w\}$ is added to E_0.
- Step 4: Go to step 2.

Solution question 124

Solution hint: From the start node, find a walk that is as long as possible.

Short solution: An attempt is made to find the longest possible walk from the start node that leads as far as possible from the start node, i.e., "deep" into the tree. If the walk can no longer be extended, since it would otherwise be closed, the algorithm returns along the walk until it reaches a node from which another unvisited node can be reached. With this node it starts again into the "depth" of the graph. When no more unvisited nodes are left, the algorithm stops and a spanning tree is the result.

Detailed solution:

- Step 1: Set E_0 as the empty set and define two lists L_1 and L_2 as empty lists. An arbitrary starting node v_1 should be chosen and written to the lists L_1 and L_2.
- Step 2: If L_1 is empty, then stop; $T = (V, E_0)$ is a spanning tree. Otherwise, take the first node v from L_1.
- Step 3: If all adjacent nodes of v are in L_2, then v is deleted from L_1. Otherwise, the "smallest" adjacent node w that is not in L_2 is chosen and written to the beginning of lists L_1 and L_2. The edge $\{v, w\}$ is added to E_0.
- Step 4: Go to step 2.

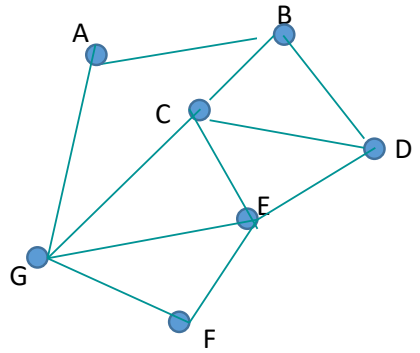
Question 125

Level of difficulty: D

Give a spanning tree for the graph in each case.

Start with node A.

a) By using the breadth-first search
b) By using the depth-first search



Solution question 125

Solution hint: a) Starting from a node walk all possible walks of length 1 b) start from the start node and find a walk that is as long as possible

Solution: a) {{A, B}, {A, G}, {B, C}, {B, D}, {G, E}, {G, F}, b) {A, B}, {B, D}, {D, C}, {C, E}, {E, F}, {F, G}}; b) several solutions are possible

8.4 Minimum spanning tree

Question 126

Level of difficulty: E

Explain Kruskal's algorithm.

Question 127

Level of difficulty: E

Explain the problem of the most convenient connection. What else is this problem called? What is meant by a minimum spanning tree? Give an example.

Solution question 126

Solution hint: "Greedily" reach for one of the cheapest edges

Short solution: Kruskal's algorithm (1956) belongs to the greedy algorithms. Here, one always greedily grabs one of the cheapest edges.

Detailed solution: For a connected weighted graph with n nodes, Kruskal's algorithm determines a minimum spanning tree T:

- Step 1: Set T: = ∅, i: = 0.
- Step 2: Among the edges not yet visited, visit the one with the smallest weight. If it closes an edge in T, discard it; otherwise, add it to T and set i: = i + 1.
- Step 3: If i = n–1, then stop, otherwise continue with step 2.

Solution question 127

Solution hint: Minimum spanning tree

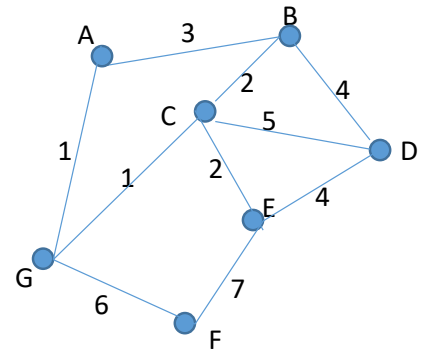
Short solution: Minimum spanning tree, connector problem, railway network with minimum total weight

Detailed solution: The problem of the most favorable connection (connector problem) is represented graph theoretically by a minimum spanning tree. A spanning tree for a weighted graph whose total weight is minimum is called minimum spanning tree T (also called minimum spanning tree or MST). Example: Railway network; all cities are to be reached with minimum total weight.

Question 128

Level of difficulty: M

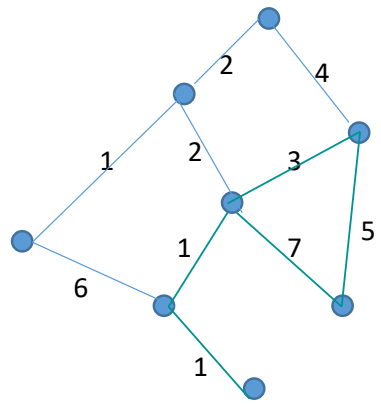
New roads are to be built between cities. The construction costs are given in millions of euros. As this sum is not available at that present moment, it is planned that each city can be reached by every other city on one of the new roads, but by only one road at a time. The construction costs should also be kept as low as possible. Therefore, a minimum spanning tree of the graph is searched.



Question 129

Level of difficulty: M

In this graph, a minimum spanning tree should be found. Specify the total weight.

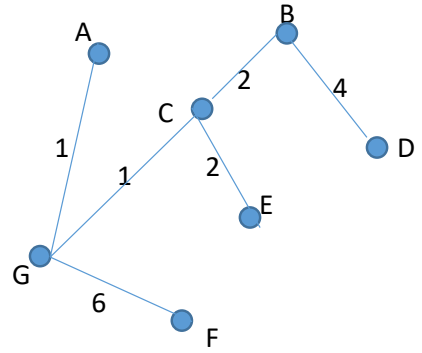


Solution question 128

Solution hint: Kruskal's algorithm

Short solution: The total weight is 16.

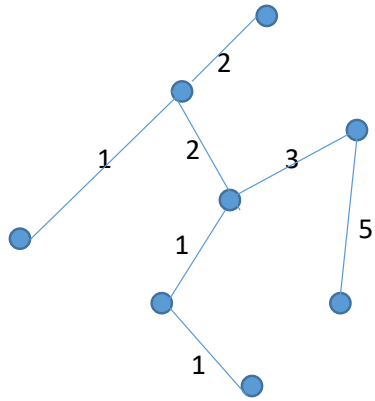
Detailed solution: To find this, Kruskal's algorithm is used and for this all weights are noted in ascending order as a list: 1 1 2 2 3 4 4 5 6 7. Then, the most favorable road is built first, followed by the next most favorable and so on. With equal weights it does not matter which one is chosen. Only care must be taken that no closed walks will be created. If all cities have a road connection, the task is solved. Total weight: 16.



Solution question 129

Solution hint: A tree containing all nodes and where the total weight is minimal.

Short solution: The total weight is 15.



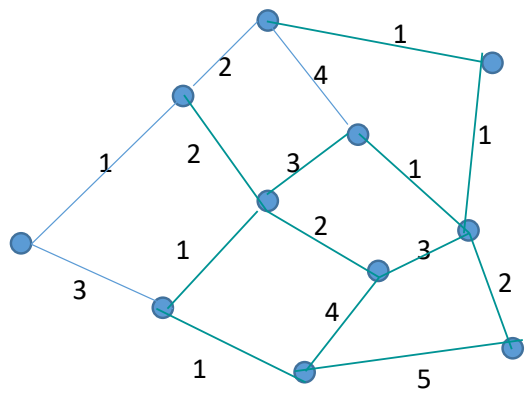
Detailed solution: 1 1 1 2 2 3 4 5 6 7

Total weight: 15.

Task 130

Level of difficulty: D

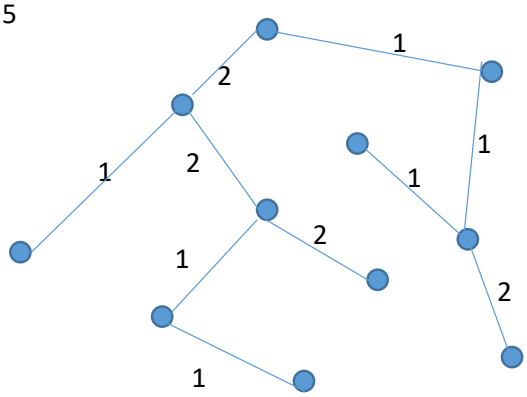
In this graph, a minimum spanning tree can be found. Enter the total weight.



Solution Task 130

Solution hint: A tree containing all nodes and where the total weight is minimal

Short solution: The total weight is 14.

Detailed solution: 1 1 1 1 1 1 2 2 2 2 3 3 3 4 4 5

Total weight: 14.